

Design and extreme structural analysis of wind turbine blades: Beam and shell model comparison and discussion for a 10-MW reference turbine

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ARTICLE INFO

Keywords:

Wind turbine blade
Composite shell
Composite beam
Nonlinear FEM
Trailing edge buckling
Nonlinear buckling
Vibration method
Imperfection-sensitivity

ABSTRACT

The progressive growth of wind turbine blades requires lightweighting to ensure aerodynamic performance. However, gaps in the comprehension of failure mechanisms, such as trailing edge buckling, lead to non-scalable experimental-based optimization frameworks. Nonlinear finite element methodologies are now central in blade design, giving insight into the structural behavior and speeding up design iteration. This work aims to examine finite element techniques from the standpoint of the DTU 10-MW reference wind turbine. The geometrically nonlinear anisotropic beam model is verified against its corresponding layered shell model for the reference turbine, followed by a comparison between linear and nonlinear stability methodologies for buckling strength assessments. The discussion on stability highlights the eminence of including nonlinear methods to account for shells' typical imperfection sensitivity in blade design routines.

1. Introduction

The decarbonization of the energy sector is a worldwide commitment to a climate-safe future, gathering several signatories for the Paris Agreement in 2015 [1]. However, global energy demand continues to increase, headed by developed countries, which requires a multidisciplinary effort to tackle policy, capacity, and technology. The Global Wind Energy Council reported a 12.4% growth in global installed wind power capacity in 2022, compared to 2020, surpassing 837 GW [2]. Offshore wind generation delivered 22% of total wind capacity growth in 2021, with an increasing share trend in the following years [3].

The modern three-bladed horizontal-axis wind turbine design (HAWT) is nowadays a metonymy of wind power generation. In HAWT, rotor diameter is a key driver of power capacity since it is proportional to the square of the blade span. Blade design optimization has thus become a core task in its conception, especially in the offshore segment with no turbine size restriction. Notwithstanding, the progressive sizing accompanies a weight trade-off, imposing lightweighting in novel blade concepts [4].

Laminated fiber-reinforced composite materials are a cost-effective blade manufacturing alternative, providing shape flexibility and high specific strength. Wind turbine blades are now slender and flexible, presenting large deformations even at service load levels [5]. Finite

element geometrically nonlinear structural analysis has thus become a rule in HAWT blade design routines [6]. Most aeroelastic codes [7–9] handle geometric nonlinearity via three-dimensional anisotropic finite element beam models, capturing large deformation, material-induced and aeroelastic couplings [10,11]. The convenience of adopting geometrically exact beam models for general optimization goals is enormous, as these models are both computationally light and accurate. However, higher hierarchy models, such as 3D geometrically nonlinear finite element shells [5,6], may be required to detect local phenomena.

Researchers have warned about local buckling in blade collapse in experimental and numerical studies since the early 2000s [12]. The trailing edge is especially prone to instability and damage due to excessive stress concentration and adhesive joint failure [13,14]. However, more than classical approaches, such as linear buckling analysis and ad-hoc material failure criteria, are needed to determine the design of the blades [15,16]. Other geometrically nonlinear sectional effects, such as shear distortion and the ovalization of hollow thin-walled structures under bending, also known as Brazier behavior, must be adequately characterized [4,15,17,18].

Understanding gaps in the damage evolution of wind turbine blades restrain ultimate failure-based design, requiring the execution of a full-scale blade certification test [12] to evaluate its integrity under

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<https://doi.org/10.1016/j.engstruct.2025.120155>

Received 24 October 2023; Received in revised form 18 February 2025; Accepted 15 March 2025

Available online 31 March 2025

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its most severe design load conditions [19]. The blade certification standard accounts for static load tests, fatigue tests, and others to assess overall blade properties [19]. However, full-scale tests are too costly and limited to simple load cases, not representing a realistic load scenario [4,13]. Finite element analysis can handle geometric nonlinearity and is a cost-effective alternative to simulate multiple load cases and give insight into elastic failure mechanisms.

A series of numerical-experimental studies aimed to gauge finite element modeling reliability for final design calculation [12,16,20–24]. Haselbach [22] raises the relevance of trailing edge adhesive connection stiffness modeling on the accuracy of the numerical results compared to experimental data and recommends a combined shell-solid modeling approach for detailed ultimate strength assessments of rotor blades. Rosemeier et al. [16] question even the current wind turbine blade design standards and certification practices concerning the adoption of linear FE methods against nonlinear methods, specifically on ultimate strength and stability limit state analyses. Shi et al. [23] investigate the influence of wind and rotational speed on the vibration characteristics of a 5 MW offshore wind turbine blade using a bidirectional fluid–structure interaction (BFSI) model, validating numerical results against wind tunnel test data.

The present work aims to explore the limits of conventional numerical methodologies for blade design, specifically applied to an offshore-sized, slender wind turbine blade. To the authors' knowledge, no previous studies have conducted a detailed structural comparison between beam and shell models for such a long and slender blade. Furthermore, this work reconnects to structural stability theory, adding a theoretical layer to previous studies correlating vibration characteristics and stress response, such as [23–27]. By incorporating the “vibration method”, this study introduces a feasible, reliable, and theoretically sound tool for performing nonlinear buckling strength analyses of an offshore-size wind turbine blade under operational loads.

The DTU 10-MW reference wind turbine was developed by the Wind Energy Department at the Technical University of Denmark to benchmark future offshore HAWT designs [28]. The turbine was considered for this work due to its offshore-compatible size and well-documented structural models. Concerning material and structural data, the reference turbine online repository [29] presents the turbine definition report [28], an ABAQUS [30] finite element layered shell model with the proposed shell-solid trailing edge approach and corresponding cross-sectional parameters for a finite element anisotropic beam model, preprocessed with BECAS [31] and further employed in this work as input to the geometrically exact Timoshenko beam model available in finite element code Giraffe [32].

The Giraffe software has demonstrated the ability to describe the kinematics of various structures. For example, in [33], a comparison was made between the Giraffe shell model and LS-Dyna [34]. In the offshore context, several studies have been conducted showcasing Giraffe's capabilities, such as in [35], where the dynamics of risers were analyzed, and the results were in agreement with those of the commercial software Orcaflex [36]. Regarding wind turbines, Lapa et al. [37] compared Giraffe's beam model results with the OpenFAST software [7], demonstrating that Giraffe can accurately capture the aeroelastic dynamic behavior of large wind turbine blades.

The first part of this work is dedicated to comparing of the anisotropic geometrically exact beam model and the nonlinear layered shell model. A modal analysis is conducted to verify the overall equivalence between both models' mass and stiffness distributions and to benchmark them against the results presented in the turbine's definition report [28]. Then, the models are compared in 12 quasi-static simulations, combined in flapwise and edgewise directions, as suggested by certification standards [19]. A Pressure side Towards Suction side (PTS) and a Leading edge Towards Trailing edge (LTT) certification load cases are discussed in detail for two load levels: a design-like load, and an extremely unrealistic higher load scenario, aiming to evaluate the applicability limits of the beam model.

Table 1

Apparent mechanical properties of the multidirectional plies [28].

Property	Uniaxial	Biaxial	Triaxial	Unit
E_1	41.63	13.92	21.79	GPa
E_2	14.93	13.92	14.67	GPa
G_{12}	5.047	11.50	9.413	GPa
$G_{13} = G_{23}$	5.047	4.539	4.539	GPa
ν_{12}	0.241	0.533	0.478	–
ρ	1915.5	1845.0	1845	kg/m ³

Table 2

Mechanical properties of the balsa wood [28].

Property	Balsa direction	Value	Unit
E_1	Radial	0.050	GPa
E_2	Tangential	0.050	GPa
E_3	Axial	2.730	GPa
G_{12}	Radial-tangential	0.01667	GPa
G_{13}	Radial-axial	0.150	GPa
G_{23}	Tangential-axial	0.150	GPa
ν_{12}	Radial-tangential	0.5	–
ν_{13}	Radial-axial	0.013	–
ν_{23}	Tangential-axial	0.013	–
ρ		110	kg/m ³

In the second part of this work, the shell stability problem is briefly contextualized, and the vibration method is introduced as a nonlinear approach to handle stability in a finite element context. Vibration patterns, load–deflection curves at a control section, and adhesive stresses are monitored along a nonlinear static equilibrium path analysis of the LTT certification load. The static analysis supports the comparison between the conventional linear eigenvalue buckling and the vibration method with respect to buckling strength assessment capabilities. An illustrative imperfection is added to the blade to highlight the role of imperfection sensitivity and the nonlinear method's robustness.

Finally, the conclusion discusses the results presented in this work. It confirms the nonlinear beam model's power as a tool for blade design optimization routines by highlighting its robustness in describing the blade's motion, even under extreme loading. Also, the presented results demonstrate the vibration method effectiveness in anticipating the onset of the stiffness degradation that accompanies buckling, endorsing its adoption for stability limit state analyses. Limitations of the proposed methodologies, concerning the bend-twist stiffness modeling in the beam, compared to the shell, and the difficulties in identifying the vibration shape modes along the equilibrium path are also pointed out, with further recommendations for new work.

2. Methods

The blade comprises sandwich glass fiber-reinforced composites with a balsa wood core. The laminate consists of stacked multidirectional plies whose apparent elastic properties, estimated using micromechanics equations and the Classical Lamination Theory (CLT), are available in Table 1 [28]. Table 2 displays the balsa wood material properties [28].

The structure is a cantilever beam, tied to the blade root. It is conceived as a load-carrying box girder with two main shear webs and an aft web from the first quarter of the blade to the tip. The ABAQUS shell structural model is based on 8-node layered shell elements (S8R), added of solid 20-node hexahedra (C3D20) to represent the suction-to-pressure sides trailing edge adhesive connection. There are composite layups for 100 sections, subpartitioned radially into 11 regions, such as webs, caps, nose, and panels. The model contains approximately 35,000 elements and 104,000 nodes [28], and its mesh is depicted in Fig. 1. The corresponding model consists of Giraffe's geometrically-exact composite beam model [32], meshed into 101 nodes, and 50 quadratic elements whose cross-sectional stiffnesses and mass properties were automatically generated in BECAS [31] from the shell model

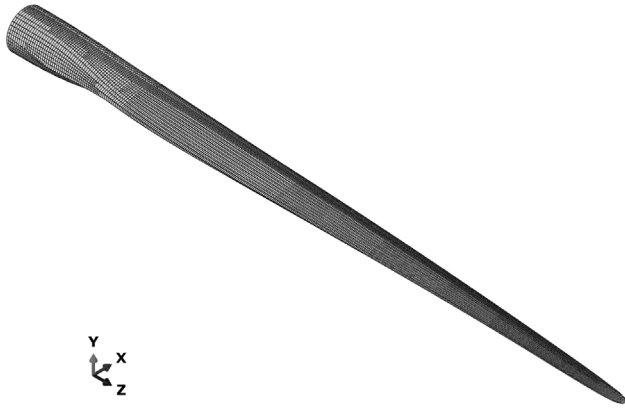


Fig. 1. Shell model finite element mesh and the rotor coordinate system.

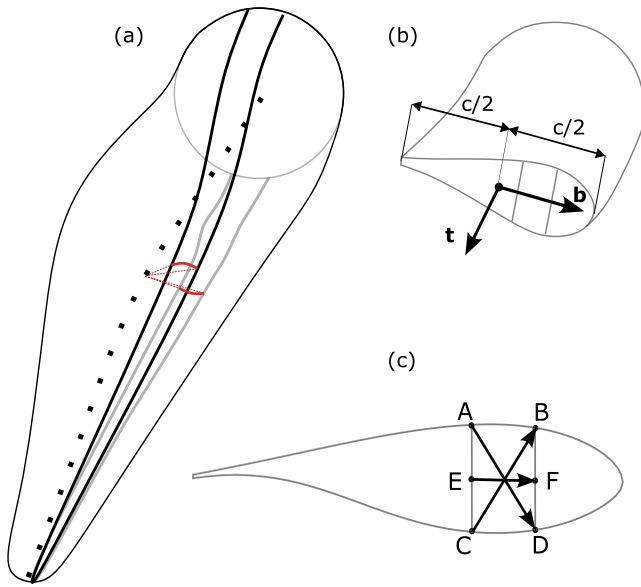


Fig. 2. The sketch above displays: (a) the shell's equivalent axis reference nodes and a kinematic distributed coupling constraint example (in red); (b) the local coordinate system, defined by the cross-sectional vectors b and t , referring respectively to bending and twist rotations, and the reference node, positioned at the semi-chord; (c) cross-sectional vectors definition: $t = \frac{AD \times CB}{\|AD \times CB\|}$ and $b = \frac{EF}{\|EF\|}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

database [28]. Complete structural and material data for both the beam and shell models are available in the turbine's online repository [29].

Neither model included blade pre-bending, which was posteriorly added to both. The prebend was added to the beam's y-coordinates, summarizing the stress-free sectional offset that characterizes the sheared blade shape. The shell pre-bending, however, required a three-dimensional geometry update, relying on the deformed shape of a previous nonlinear displacement-based static analysis.

Despite referring to the same geometry, the beam's degrees of freedom derive from the extrusion of planar cross-sections along its axis, while the shell's degrees of freedom are bound to its surface. A proper setup of matching boundary conditions and the comparison of the following results demand the generation of consistent quantities.

An alternative is to reproduce the beam's cross-sections within the shell model by creating equivalent degrees of freedom, which is done by adding the beam's axial nodes – defined at the airfoils' semi-chords – to the shell's model database, thereby replicating an equivalent shell axis. To define the cross-sections in the shell model, beam's section planes

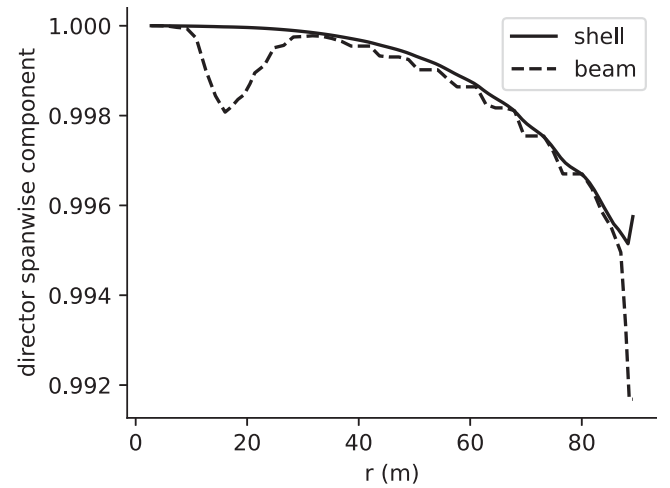


Fig. 3. The figure depicts the evolution of the director's spanwise component for both the beam and the shell models.

are intersected with the shell surface, from which two key vectors are extracted: a twist vector t – normal to the section and derived from the cross-product of vectors in the section plane – that represents in-plane twist, and a bending vector b , coplanar to the section, which accounts for out-of-plane rotations (either flapwise or edgewise). Given that the shell experiences local deformations under load, nodes located at the shear webs are selected to geometrically define these section vectors, as shown in Fig. 2.

To simulate axial loading in the shell model, the reference nodes are coupled to the spar cap nodes using kinematic coupling constraints, as defined in ABAQUS. These surface-based constraints couple the motion of a collection of nodes on the shell surface to that of a reference node [30]. The approach, illustrated in Fig. 2, generates an equivalent cross-section in the shell without restricting local deformations, and is similar to the method proposed in [21]. Additionally, coupling the nodes to the spar caps allows for the application of equivalent pointwise loads to both models, effectively simulating aerodynamic pressures in a simplified manner while avoiding unrealistic stress concentrations on the blade surface.

Although Saint Venant's principle sustains the immediate correspondence regarding forces and translations between the axes' nodes, the director vectors that define the cross-sectional planes must also be alike to assess both models' moments and rotations. Unlike the beam model, the shell is not an extrusion of sections, and since the adopted finite element model was already meshed [29], it was impossible to posteriorly idealize perfectly parallel cross-sections along the whole blade axis so that rotational quantities would be completely consistent. Fig. 3 highlights the slight difference between the spanwise components of both models' axial directors up to the first third of the blade and close to the tip, besides the discontinuities due to beam discretization.

A control section is set at $z = 80.5$ m to assess both models' displacements and rotations, chosen for capturing the overall structural behavior along the span while avoiding systematic inconsistencies in rotational quantities due to diverging director vectors.

3. Compared simulations

3.1. Modal

Free vibration analyses were performed for the beam and the shell models. Table 3 points to an excellent agreement between the natural frequencies of the first six modes for the Giraffe's finite element model and the turbine's definition report.

Table 3

First natural frequencies for Giraffe beam and DTU10MW report [28].

Giraffe f (Hz)	Report f (Hz)	Difference (%)	Mode
0.615	0.61	0.82%	1st flapwise mode
0.938	0.93	0.86%	1st edgewise mode
1.753	1.74	0.75%	2nd flapwise mode
2.776	2.76	0.58%	2nd edgewise mode
3.583	3.57	0.36%	3rd flapwise mode
5.671	5.69	−0.33%	1st torsional mode

Table 4

First natural frequencies for beam and shell models.

Beam f (Hz)	Shell f (Hz)	Difference (%)	Mode
0.615	0.611	−0.77%	1st flapwise mode
0.938	0.964	2.71%	1st edgewise mode
1.753	1.752	−0.05%	2nd flapwise mode
2.776	2.883	3.71%	2nd edgewise mode
3.583	3.577	−0.18%	3rd flapwise mode
5.671	5.707	0.62%	1st torsional mode
–	5.747	–	1st torsional mode

Table 4 depicts adherent mass and stiffness distribution between both models, although the shell is slightly stiffer in the edgewise direction. Since BECAS documentation [31] does not refer directly to local cohesive bondlines, the deviation between both models' edgewise stiffness is possibly due to the absence of adhesive trailing edge modeling in the beam.

Also, while the shell model presents two symmetric torsional modes with close natural frequencies, the first torsional mode is unique for the beam model. Local vibration modes take place for higher frequencies in the shell model.

3.2. Static

From the extreme design loads available in the turbine's report [28], 12 combined flapwise and edgewise bending simplified load cases were derived. The goal is not to precisely replicate the full load envelope of the turbine but rather to assess the capabilities of the structural models using a reference that is sufficiently representative of the design loading conditions while simple enough for replication by other researchers.

The proposed simulations are quasi-static, comprising the clamped blade loaded uniformly by pointwise forces across the axes' nodes, defined at the sections' semi-chords, displayed in Fig. 2. The semi-chords were chosen instead of the elastic centers to explore how both models respond to the coupling between bending and torsion that often arise in laminated structures. Also, a volume-proportional damping coefficient of 1×10^{-4} was set in ABAQUS automatic stabilization to account for local instability [30].

Table 5 and Fig. 4 summarize the simulated load cases for both the design and extreme load scenarios. The load direction is measured counterclockwise from the chord direction.

The beam and the shell models presented a similar structural response in all the 12 simulations, even under extreme load conditions, as shown in the detailed report of both models' displacements at the blade tip, available in Appendix. The adherence between the models indicates the nonlinear beam model's suitability in describing the overall blade's structural behavior in design routines, and a closer comparison of both models' behavior is performed next.

Two certification loads – Pressure side Towards Suction side (PTS) and Trailing edge Towards Leading edge (TTL) load cases – are discussed in detail. The proposed analyses compare cross-sectional quantities for both models along loading and exhibit cross-sectional local deformations for the shell model.

Table 5

Design and extreme load cases.

Load case	Direction deg	Design kN/m	Extreme kN/m	Ratio
1	0	8.6	35.8	416%
2	30	8.4	36.6	436%
3	60	10.6	38.5	363%
4	90	18.8	38.1	203%
5	120	10.1	33.8	337%
6	150	7.1	17.5	247%
7	180	7.0	15.0	215%
8	210	7.1	18.9	266%
9	240	6.2	37.8	610%
10	270	7.3	41.7	571%
11	300	12.4	36.8	297%
12	330	9.4	34.9	371%

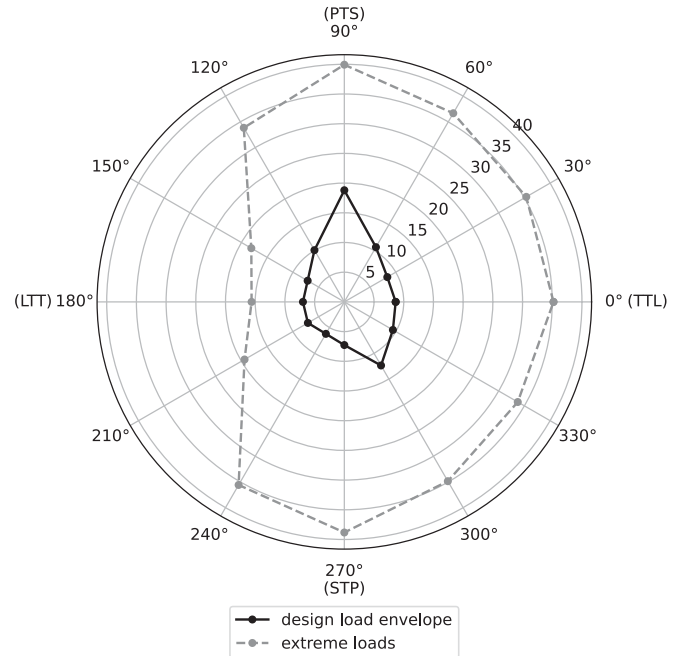


Fig. 4. Design load envelope and extreme applied distributed uniform loads for the 12 directions. The angular coordinate represents the angle from the applied force direction and the edgewise direction, and the radial coordinate, the magnitude of the distributed uniform load, in kN/m. Main certification load directions are highlighted.

3.2.1. Pressure side Towards Suction side (PTS) load case

The control section displacements and rotations for the Pressure side Towards Suction side certification simulation are presented in Table 6 for the proposed design and extreme scenarios, and Figs. 5 and 6 display these quantities' evolution between them. Axial displacements are described in the rotor coordinate system presented in Fig. 1, while rotations refer to the local coordinate system, depicted in Fig. 2, for sounder structural interpretability.

Although the beam is slightly stiffer than the shell, both models present adherent displacement results even at loads above twice the design load, with less than a 5% absolute deviation between them. Concerning the rotational degrees of freedom, the models also have a similar response in bending, but the beam presents higher twist angles. This variation, however, is not enough to affect the overall structural response since twisting is a coupling effect that derives from applying the load in the semi-chords instead of the shear centers.

It is interesting, however, that the deviations in the quantities, either translational or rotational, decrease with the loading despite the expected increasing nonlinear local sectional effects. In fact, the local deformation is significant, as shown in Fig. 7, that overlays the shell's undeformed idealized section nodes to the back rotated deformed

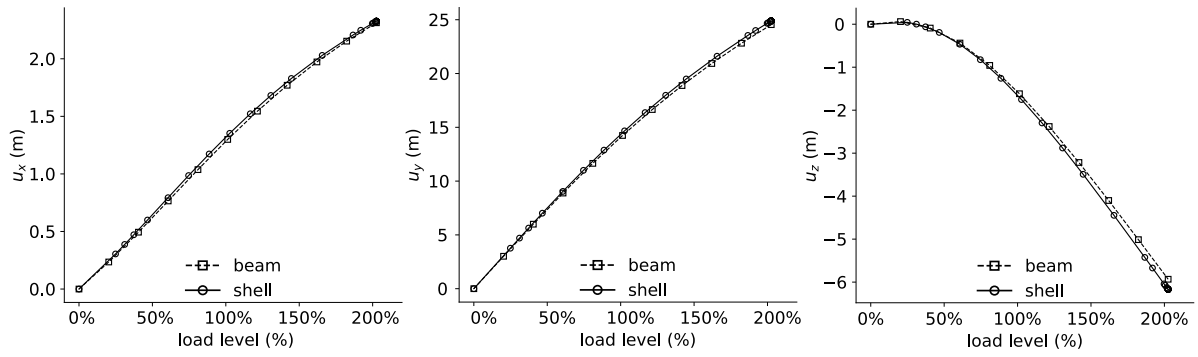
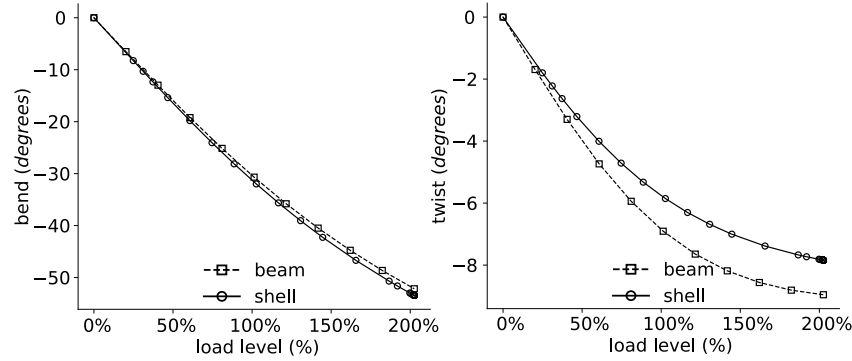
Fig. 5. Shell and beam displacements at $z = 80.5$ m for the PTS load case.Fig. 6. Shell and beam rotations at $z = 80.5$ m for the PTS load case.

Table 6

Displacements and rotations at $z = 80.5$ m for the PTS load case.

			Design	Extreme
u_x	Beam	m	1.28	2.31
	Shell	m	1.32	2.33
	Deviation	%	-2.72	-0.56
u_y	Beam	m	14.08	24.54
	Shell	m	14.34	24.89
	Deviation	%	-1.77	-1.40
u_z	Beam	m	-1.57	-5.93
	Shell	m	-1.65	-6.17
	Deviation	%	-4.99	-3.82
Bend	Beam	deg	-30.35	-52.16
	Shell	deg	-31.27	-53.44
	Deviation	%	-2.95	-2.40
Twist	Beam	deg	-6.85	-8.96
	Shell	deg	-5.76	-7.84
	Deviation	%	19.00	14.19

Table 7

Displacements and rotations at $z = 80.5$ m for the LTT load case.

			Design	Extreme
u_x	Beam	m	-2.16	-4.66
	Shell	m	-2.08	-4.71
	Deviation	%	4.00	-1.11
u_y	Beam	m	-0.43	-0.98
	Shell	m	-0.45	-1.02
	Deviation	%	-5.01	-4.47
u_z	Beam	m	-0.09	-0.31
	Shell	m	-0.08	-0.32
	Deviation	%	8.53	-1.86
Bend	Beam	deg	3.81	8.25
	Shell	deg	3.53	8.23
	Deviation	%	7.87	0.23
Twist	Beam	deg	-0.89	-2.19
	Shell	deg	-0.59	-1.62
	Deviation	%	50.63	35.31

section nodes, isolating the local section deformation in the proposed scenarios. Transverse shear distortion and out-of-plane cap flattening caused by the Brazier effect [4,18] – section ovalization due to bending – are present at both design and extreme load conditions. Nonetheless, as global behavior dominates over local effects, the softening induced by the latter does not significantly depart between the two models.

This behavior can be partially explained by the relationship between section height and bending stiffness. In the case of flapwise bending, the rotation of the blade section due to applied loads increases the section height, as the deformed sections are longer chordwise. The twist results in a moment of inertia increase, which counteracts the softening effects of local deformations, such as section ovalization. As a result, the shell sections become stiffer with increasing load, and the bending response becomes less sensitive to local deformations that would otherwise reduce the section's bending stiffness.

Additionally, the sectional twist reduces the lever arm between the semi-chords' and shear centers' projections. As the shear center is, by

definition, the point in the section where an applied external shear force does not induce twisting, enclosing it to the load application point reduces incremental twisting.

In summary, while local effects such as shear distortion and ovalization soften the structure under load, global rotation kinematics diminishes the overall impact of these local effects, explaining why the translational and rotational deviations decrease with increasing load, despite the expected increase in local deformations.

3.2.2. Leading edge Towards Trailing edge (LTT) load case

The control section results for the Leading edge Towards the Trailing edge certification load are presented in Table 7. Fig. 8 depicts the axial displacement evolution between the proposed scenarios in the rotor coordinate system, and Fig. 9, the development of twisting and bending rotations.

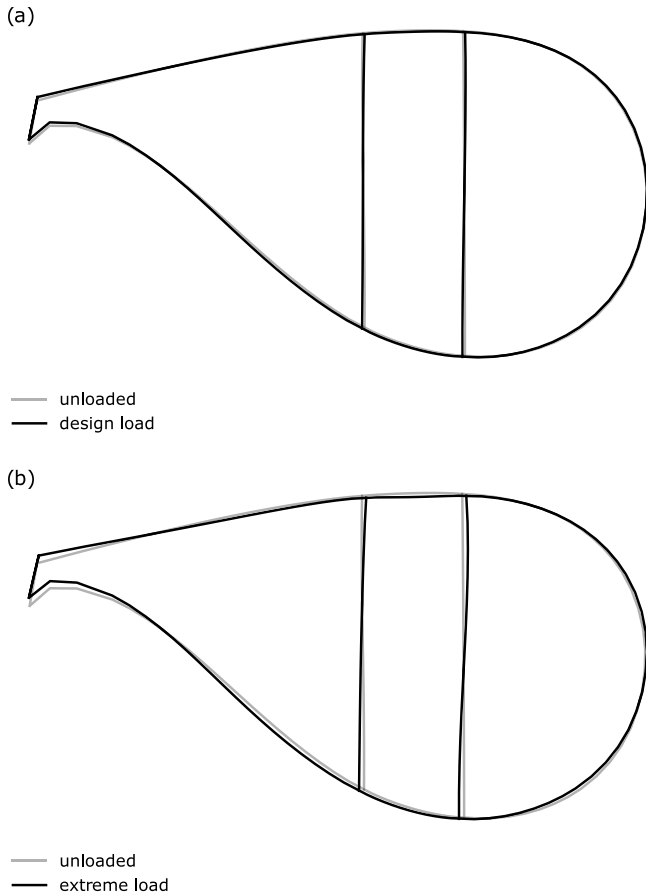


Fig. 7. Section local deformation at $z = 19.21$ m for (a) design load level and (b) extreme load level for the PTS load case.

The models have a good agreement for displacements and bending rotations and present linear behavior in flapwise and edgewise displacements up to approximately 85% of the extreme load level, from which the shell exhibits nonlinear behavior. This nonlinearity represents the onset of the trailing edge buckling, shown in Fig. 10.

The diverging bend-twist coupling stiffnesses are also observed in this simulation, although the coupling effect is even less representative than in the PTS load case. The bend-twist coupling is reduced due to closer semi-chords' and shear centers' projections in the edgewise direction, which lowers the product of inertia that causes the coupling effect. As already observed in the modal analysis 3.1, the shell is slightly stiffer than the beam in the edgewise direction. The almost linear bending behavior observed in Figs. 8 and 9 is due to the blade's high

edgewise stiffness, which can be perceived from lower displacements and rotations compared to the previous case and minimal section local deformation until the onset of buckling. Notwithstanding, the airfoil-shape blade section gets severely distorted after buckling, as outlined in Fig. 11.

The importance of accurately modeling the adhesive bond between the pressure and suction sides via solid elements is noteworthy, becoming even more relevant after buckling, as summarized in Figs. 12 and 13, which present the out-of-plane stresses captured by the solid elements and their ratio concerning the maximum principal stresses at the buckled trailing edge (σ_{33}/σ_{max}). The absence of these elements would underpredict the trailing edge's stiffness response, as already pointed out in [22], since the shell structural elements cannot inherently represent out-of-plane stress.

Edgewise gravity accompanies blade enlarging, majoring loads, and trailing edge buckling liability [4]. Addressing the phenomenon is essential for an ultimate strength assessment of the blade, for buckling not only softens the structure but might also trigger progressive damage mechanisms such as trailing edge bondline debonding and sandwich core failure [21,22]. Buckling capacity is usually assessed via either linear eigenvalue buckling, or nonlinear geometric finite element analysis [4], and both approaches are discussed for the stability analysis of the LTT load case in the next section.

4. Stability analysis

Experimental full-scale test data present trailing edge buckling at early load stages, even though blades are designed to prevent local phenomena [12,21,38]. Test data also show that the energy release rate increase in the trailing edge joint that indicates the onset of progressive damage accompanies the trailing edge buckling wave formation, pointing to the phenomenon as a primary failure mechanism [38].

Although assessing the buckling strength of a wind turbine blade is conventionally a synonym for obtaining critical loads via linear eigenvalue buckling analysis [39], linear methods may lead to unsafe designs, severely overestimating critical loads in shell structures [4, 40,41]. These structures are notoriously imperfection-sensitive, buckling at unexpectedly early loading levels due to the critical load decrease caused by the strongly nonlinear behavior they exhibit in the post-buckling regime – the equilibrium states that follow an initial bifurcation point or limit point under increasing load, where the structure deforms significantly beyond the pre-buckling configuration [41, 42]. Simultaneous mode design – structural optimization with coincident or very close buckling modes – can harshen imperfection sensitivity's role, giving rise to the unexpected nonlinear coupling of buckling modes and eroding the benefits of a nominally optimum design [42]. Therefore, blade design optimization frameworks must evaluate all the post-buckling equilibrium paths instead of solely classical critical loads and include the effects of unavoidable manufacturing imperfections [42].

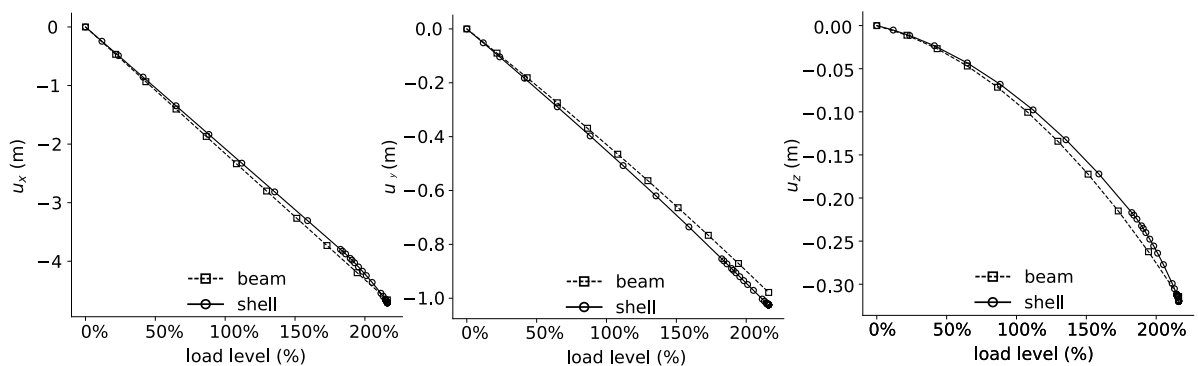


Fig. 8. Shell and beam displacements at $z = 80.5$ m for the LTT load case.

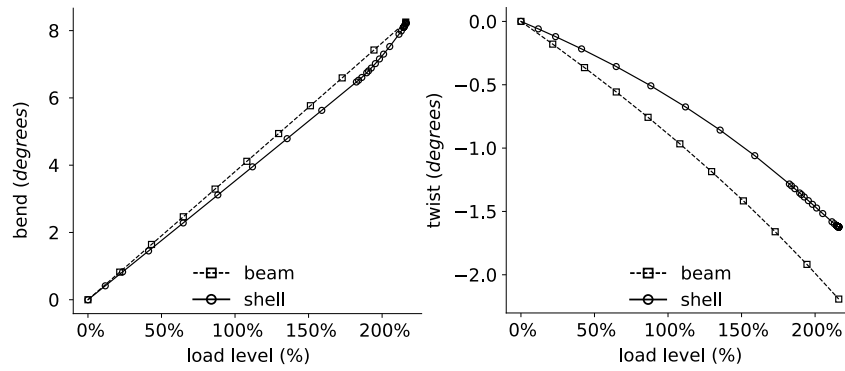


Fig. 9. Shell and beam rotations at $z = 80.5$ m for the LTT load case.

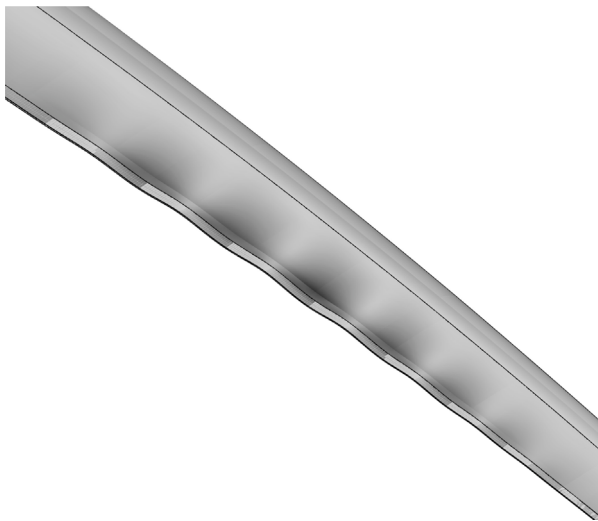


Fig. 10. Trailing edge buckling.

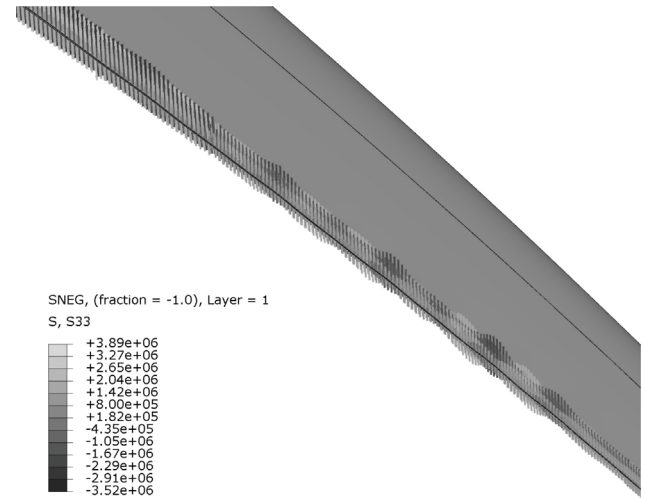


Fig. 12. Out-of-plane stress (σ_{33}) after buckling (85% of extreme load).

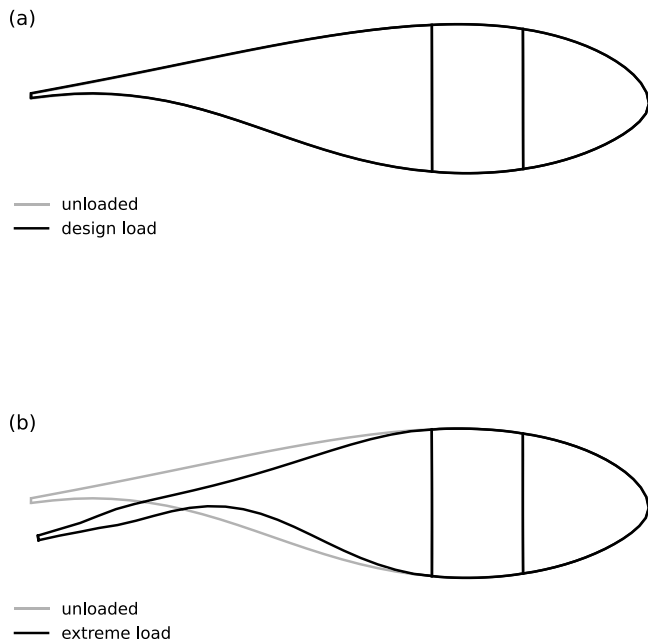


Fig. 11. Section local deformation at $z = 60.67$ m for (a) design load level and (b) extreme load level for the LTT load case.

A nonlinear approach for evaluating stability along the equilibrium path is theoretically based on what Ziegler [43] called the “vibration method”, which consists of obtaining vibration modes for configurations in both the pre and post-buckling branches. Several studies employ modern finite element techniques not only to validate the relationship between the buckling load and the variation of natural frequencies along the equilibrium path in real engineering structures, such as thin plates and cylindrical shells, but also to predict buckling loads via a nondestructive method — an area of study known as vibration correlation techniques (VCT) [25–27]. Tracking changes in natural frequencies along the equilibrium path enables the detection of bifurcation and limit points, along with their corresponding buckling modes [44]. These modes are determined based on the updated stress state and boundary conditions of the structure throughout the finite element load increments. Convergence problems in tracing the post-critical path using classical load-step methods can be circumvented by adding artificial damping, preventing singularities in the tangent stiffness matrix caused by local phenomena [45].

The scheme is motivated by Lyapunov’s concept of kinetic instability that infers the stability of a configuration from the evolution of the oscillations derived from a small perturbation of the equilibrium [43]. Choosing a kinetic stability criterion is required to capture the dynamic behavior expected in shell instability, such as mode interaction and mode jumping, the sudden jump in natural frequencies due to buckle pattern change [41,46]. Lyapunov’s theorems state that the stability of a nonlinear system can be determined from the roots of its characteristic linearized motion equation, ω^2 [43], which translates the problem into the abovementioned free vibration analysis, with natural

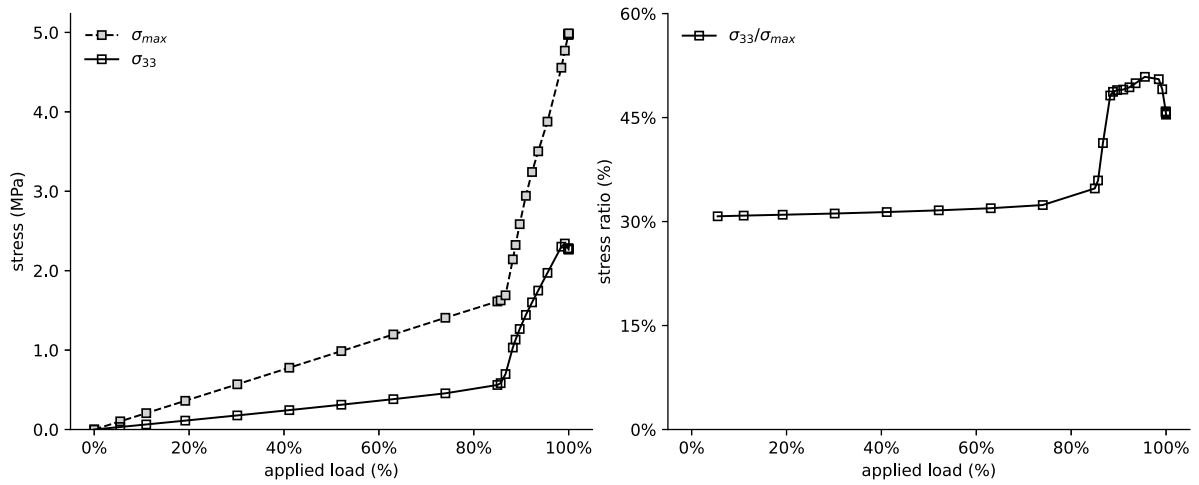


Fig. 13. (left) Out-of-plane and maximum principal stresses, respectively σ_{33} and σ_{max} , and (right) their ratio σ_{33}/σ_{max} vs. loading at a trailing edge buckled section ($z = 59.80$ m).

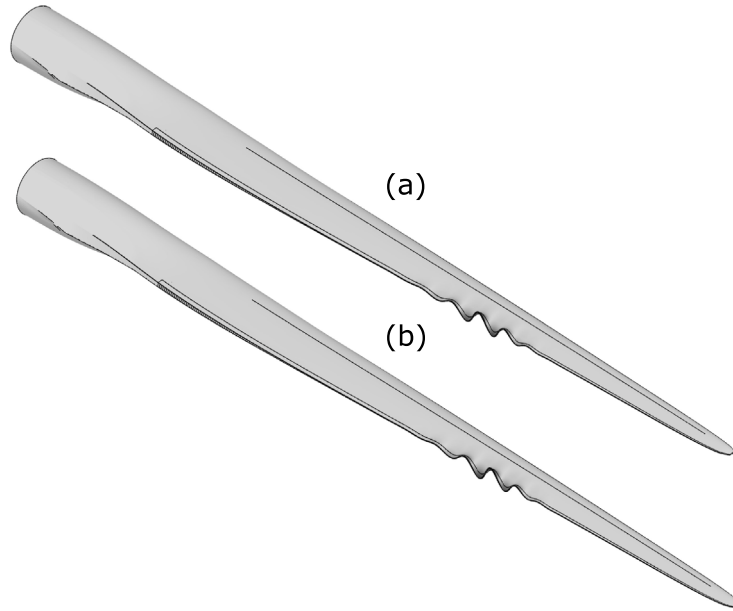


Fig. 14. First (a) and second (b) linear buckling modes, with critical loads of magnitudes 12.92 kN/m, and 12.93 kN/m.

frequencies $f = \sqrt{\omega^2}/(2\pi)$. The static equilibrium is stable if the eigenvalues' real part is positive and unstable if negative, revealing whether a bifurcation or a limit point when it drops to zero. In summary, examining a perfect structure's frequency spectrum allows for detecting the buckling mode by observing the vanishing of a particular frequency at its corresponding critical load [47]. Since natural frequencies relate directly to the eigenvalues of the structure's stiffness matrix, in imperfect structures, the frequency spectrum exhibits a gradual, rather than sudden, behavior that reflects the loss of stiffness within the post-buckling regime. Even though natural frequencies do not necessarily tend to zero in these structures, experimental studies have noticed a significant decrease in the first natural frequencies near the buckling load [25].

4.1. Perfect blade stability

The previous Section 3.2.2 presented trailing edge buckling in the DTU 10-MW perfect shell model under the extreme LTT load condition. A linear eigenvalue buckling analysis with a 50% design preload was performed to reproduce the buckling analysis reported in [28]. Fig. 14

shows almost coincident trailing edge buckling loads, indicating the possibility of the occurrence of mode interaction in the post-buckling equilibrium.

Fig. 15 shows the evolution of the non-dimensional eigenvalues, $\beta = \omega^2/\omega_0^2$, along the nonlinear equilibrium path. The first critical load of the perfect model is reached at approximately 85% of the extreme load, at $P_{crit} = 12.69$ kN/m, when the trailing edge local mode eigenvalue drops to zero. The first critical load attainment is also marked by three simultaneous nonlinear effects: the onset of nonlinearity in the control section's load-deflection curve (Figs. 16 and 17), the bifurcation in the principal stresses in the trailing edge bondline (Figs. 18 and 19), and the mode interaction. The latter can be identified by the formation of a trailing edge wave vibration pattern in the first two edgewise modes, depicted in Fig. 20, and the respective drop in these modes' eigenvalues, in Fig. 15. Figs. 18 and 19 display the principal stresses for a buckled section trailing edge element at $z = 59.80$ m. The increase in tensile stresses that characterize peeling is noteworthy and probably anticipates the initiation of a progressive damage mechanism of trailing edge bondline debonding [21,22].

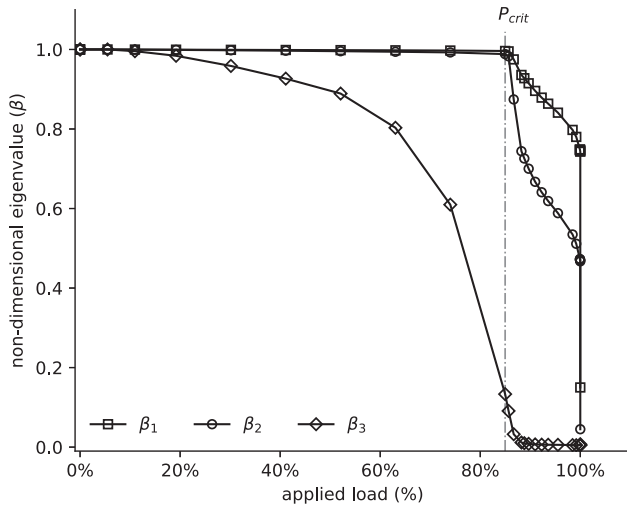


Fig. 15. Non-dimensional eigenvalues ($\beta = \omega^2/\omega_0^2$) vs. loading. Modes 1 and 2 correspond to the first two edgewise modes, presented in 3.1, and 3, to the trailing edge local mode. The dotted vertical line highlights the mode interaction effect at the critical load.

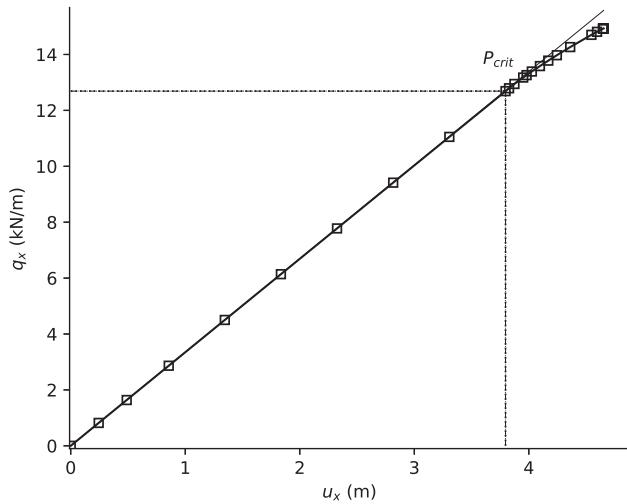


Fig. 16. Load-deflection curve for the control section ($z = 80.5$ m). The first equilibrium path's extrapolation segment highlights the onset of nonlinearity.

The similarity between the critical loads obtained from the vibration method, $P_{crit} = 12.69$ kN/m, and the linear eigenvalue buckling analysis, $P_{crit}^{lin} = 12.92$ kN/m, is not unexpected, given the linear behavior presented by the blade in the LTT load case before the onset of trailing edge buckling, shown in the 3.2.2, specifically in Figs. 8 and 9.

However, since shells are notably imperfection-sensitive, and non-linear phenomena such as mode interaction occur, adding geometric perturbations to the model likely reduces the nonlinear critical load, pointing to an overly optimistic buckling strength estimate. Aside, it is worth mentioning that although the perfect structure does not buckle at design load levels, the partial safety factors attained in both linear and nonlinear analyses are probably lower than $\gamma_{MD} = 2.042$, reported in [28]. The report evaluates the blade's buckling capacity for 27 cross sections measuring approximately 1.2 times the local chord length [28]. Yet, the trailing edge longitudinal wave observed in the complete blade finite element model comprises three of the 27 cross-sections analyzed in the report, with a length of approximately 15.5 m,

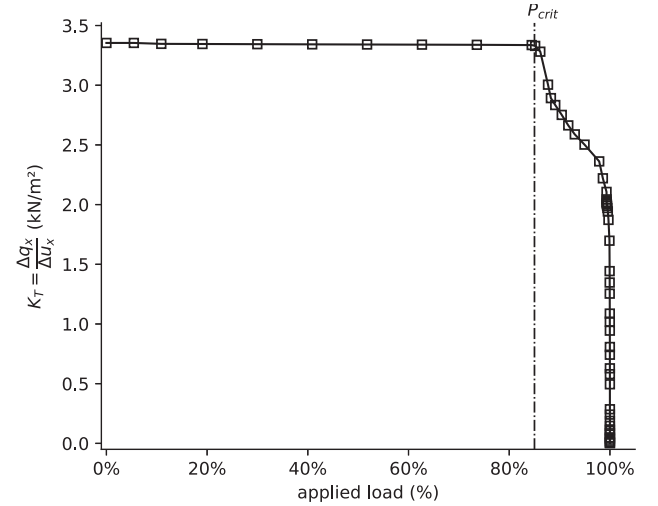


Fig. 17. Control section's ($z = 80.5$ m) tangent stiffness ($K_T = \Delta q_x/\Delta u_x$).

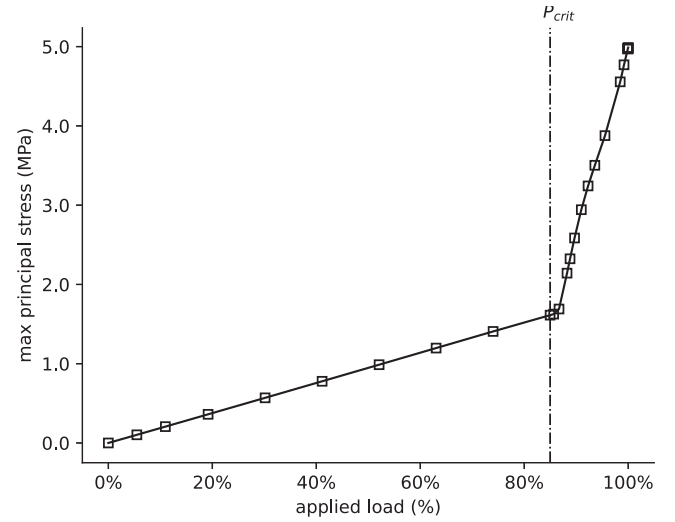


Fig. 18. Maximum principal stress vs. loading at the trailing edge of a buckled section ($z = 59.80$ m).

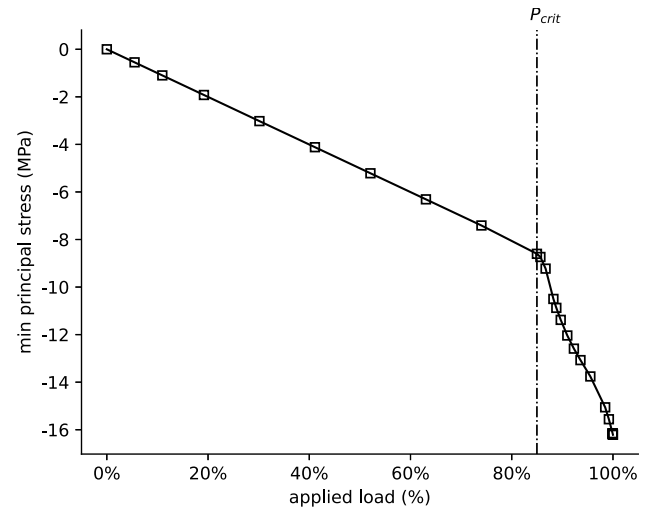


Fig. 19. Minimum principal stress vs. loading at the trailing edge of a buckled section ($z = 59.80$ m).

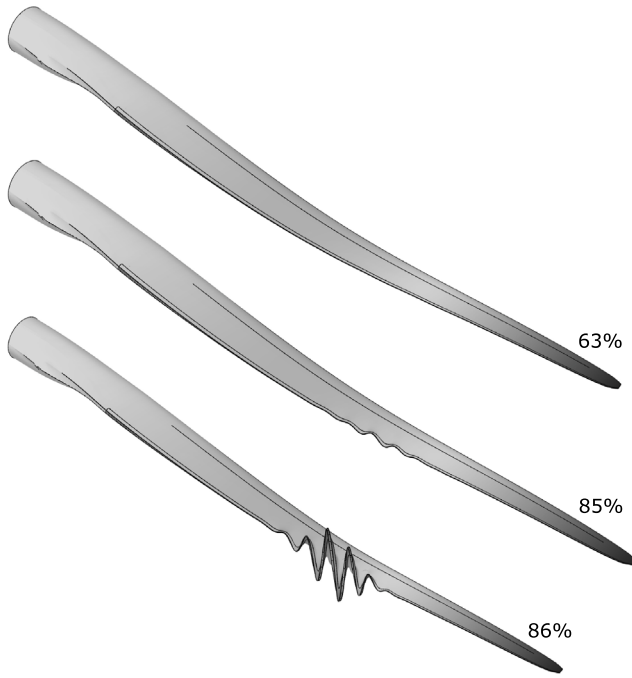


Fig. 20. Mode interaction in the second edgewise mode (β_2) for three increasing load levels. The trailing edge buckling wave can be seen at 85% of the extreme load.

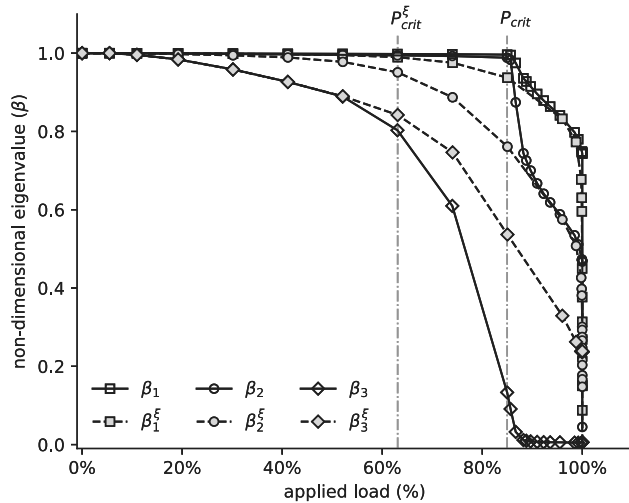


Fig. 21. Non-dimensional eigenvalues ($\beta = \omega^2/\omega_0^2$) vs. loading for the perfect and imperfect (dashed) blades. Drops in the edgewise global modes occur earlier in the imperfect structure.

which indicates that the cross-sections' slenderness ratio may also play a role in buckling strength assessments.

4.2. Imperfect blade stability

Wind turbine blades undergo a multi-step manufacturing process that inevitably introduces imperfections, whether during mold preparation, composite layup, curing, or bonding. The first buckling mode represents the most sensitive deformation shape, meaning that small imperfections of this form have the most significant impact on reducing the buckling load. To account for this effect, a geometric imperfection

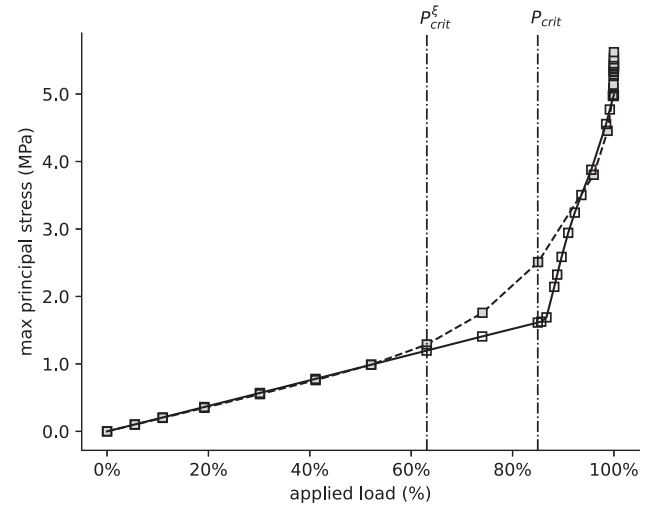


Fig. 22. Maximum principal stress vs. loading at the trailing edge of a buckled section ($z = 59.80$ m) for the perfect and imperfect (dashed) blades.

shaped according to the first linear buckling mode, and a magnitude of $\xi = 8.75$ mm, was applied to the shell surface to create an imperfect structure for subsequent critical load evaluation.¹ The definition of the critical load of the imperfect structure is not as distinguishable as the perfect's. In the first, the monitored quantities, whether stresses or displacements, tend to increase smoothly rather than bifurcate sharply, as in the latter. The imperfection also smoothens the eigenvalue spectrum in the vicinity of the buckling load [27], requiring the evaluation of stiffnesses and displacements along the nonlinear equilibrium path to detect local phenomena instead of directly monitoring vanishing eigenvalues. Notwithstanding, the vibration equilibrium path can still serve as a useful proxy of the local stiffnesses, offering a more exhaustive pattern-based viewpoint of the structural behavior instead of a single arbitrarily chosen probe.

Although the definition of the critical load for imperfect structures is debatable, monitoring the trailing edge buckling wave remains essential in wind turbine blade design, as it is a primary failure mechanism. Thus, the critical load can be generically defined as an engineering threshold — the load level beyond which undesirable effects, such as large stresses, large strains or another engineering criterion that would represent a damage in the structure. In practice, this implies tracking a key variable along the nonlinear equilibrium path for both perfect and imperfect structures, using the first as a theoretical benchmark. This threshold can be loosely defined as the loading level beyond which the imperfect structure's probe variable diverges from the corresponding pre-buckling behavior of the perfect model or, when applicable, by ad-hoc criteria such as maximum strains or stresses.

In the trailing edge stability problem, the maximum principal stress that characterize peeling — directly correlated with the failure mechanism — emerges as a strong candidate for a probe variable. Nevertheless, it is noteworthy that, as in the perfect model, the nonlinear effects following the principal stresses' bifurcation — the drop in the local trailing edge mode eigenvalue and its interaction with the first edgewise modes (Fig. 21), along with the overall stiffness decrease (Figs. 24 and 25) — also follow the smooth divergence between the imperfect structure principal stresses and the perfect's, shown in Figs. 22 and 23. The load level at which these nonlinear effects emerge, $P_{crit}^\xi = 9.41$ kN/m, can be defined as critical, at a 25.9% lower buckling load than the perfect model's, $P_{crit} = 12.69$ kN/m.

¹ The imperfection scaling factor was set as 1/400 of the critical buckle wavelength, as recommended by certification standards [48].

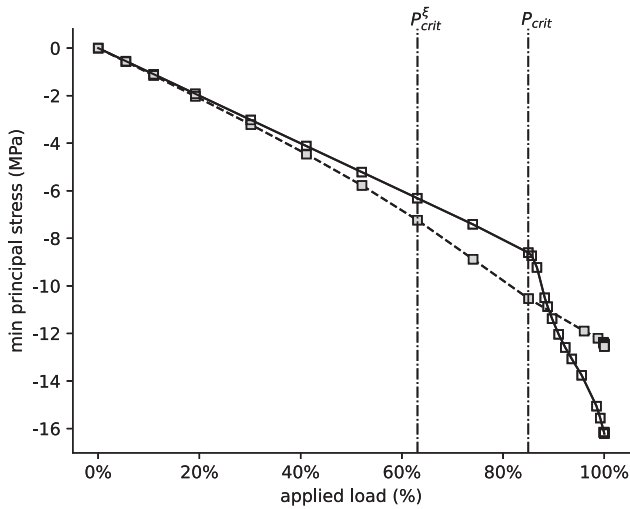


Fig. 23. Minimum principal stress vs. loading at the trailing edge of a buckled section ($z = 59.80$ m) for the perfect and imperfect (dashed) blades.

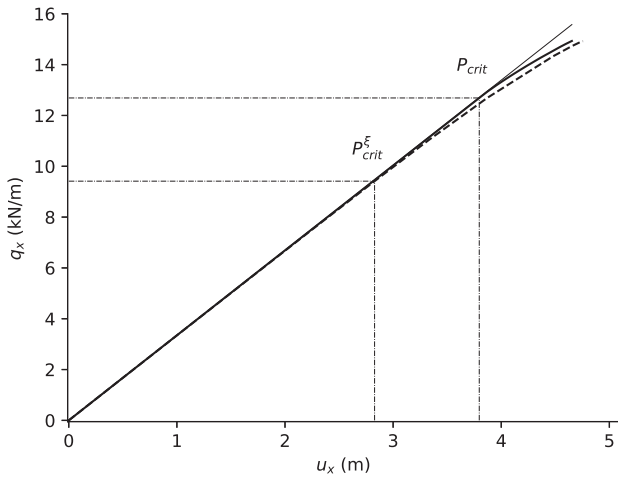


Fig. 24. Load-deflection curve for the control section ($z = 80.5$ m). The first equilibrium path extrapolation segment highlights the onset of nonlinearity in the imperfect blade.

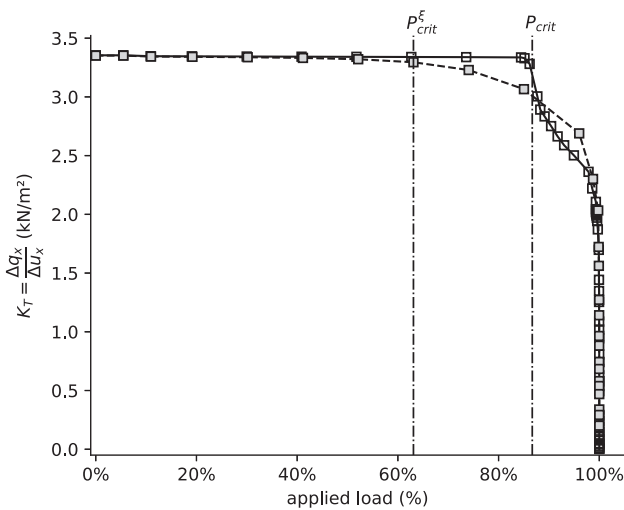


Fig. 25. Control section ($z = 80.5$ m) tangent stiffness ($K_T = \Delta q_x / \Delta u_x$) for the perfect and imperfect (dashed) blades. The reduction in the tangent stiffness is much smoother in the imperfect blade.

Although illustrative, this example shows how analyzing the eigenvalue spectrum can anticipate local phenomena and help select physically meaningful probe variables in imperfect structures. Also, it points out how small manufacturing imperfections can affect wind turbine blades' buckling strength assessment, which requires evaluating the stability along the structure's nonlinear equilibrium path. Yet, for an actual buckling capacity assessment, the shape and magnitude of a real blade's imperfections must be adequately characterized, and an imperfection-sensitivity study must be carried out.

5. Conclusions

This work examined the DTU 10-MW reference wind turbine from the standpoint of two conventional wind turbine blade design finite element methodologies: geometrically nonlinear anisotropic beam modeling and shell linear buckling analysis.

One of the key findings is the verification of the nonlinear geometrically exact Timoshenko beam model as a powerful tool for offshore-sized wind turbine blade design optimization. The beam model accurately describes the overall structural behavior, even under extreme load conditions, when compared to the shell model. However, there is room for improvement in using the beam model for bend-twist coupling optimizations, as it exhibited slightly softer behavior in the edgewise direction, likely due to the absence of the adhesive trailing edge bond, which is included in the shell model.

Notwithstanding the beam model's precise description of the structure's global behavior, finite element nonlinear layered shell models are also required in design routines to assess buckling strength, for local phenomena such as trailing edge buckling may trigger progressive damage, consisting of a primary failure mechanism.

Rather than predicting final trailing edge failure, this work aimed to anticipate trailing edge buckling. Accounting for damage in design routines is often non-trivial and computationally expensive as failure modes' coupling results in a highly complex and nonlinear behavior [38]. Since fracture initiation is closely related to buckling onset, a more reliable final blade failure modeling strategy would require modeling damage evolution via three-dimensional failure criteria, as presented in [49].

Alternatively, structural safety can be ensured by simply avoiding the complex fracture and damage phenomena that follow the buckling load. Despite commonly employed, linear eigenvalue methods may lead to non-conservative estimates of buckling strength for shell structures, typically imperfection-sensitive, leading wind turbine blades to buckle at unexpectedly early loading levels. Therefore, blade design optimization frameworks must incorporate manufacturing imperfections and assess the stability equilibrium path using nonlinear methods. The vibration method, demonstrated to be both feasible and reliable in this work, is a valuable tool in this regard.

However, limitations remain, particularly in the extensive effort required to accurately characterize multiple manufacturing imperfections for a reliable estimate of buckling strength. Additionally, identifying corresponding mode shapes along the equilibrium path, especially early on when local vibration modes couple with other modes, can be challenging. Although the process becomes clearer as the critical load is approached, it remains a non-trivial task. This is particularly true for structures with complex geometry and material properties, and there is room for further development in future work.

This work aims to explore the limitations of traditional modeling approaches, offering a nonlinear alternative to improve the accuracy of buckling strength predictions, particularly in the presence of manufacturing imperfections that are characteristic of real-world structures. By providing a method to anticipate trailing edge buckling, this study contributes to the development of more reliable finite element-based design frameworks, helping to avoid premature failures that can arise from overly optimistic buckling strength predictions. Ultimately, it highlights the need for further studies to better understand the sensitivity of wind turbine blades to imperfections, which is crucial not only for optimizing structural designs but also for refining manufacturing standards, thereby enhancing turbine efficiency and reliability.

CRediT authorship contribution statement

Isadora Toledo de Almeida: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Gabriel Vicentin Pereira Lapa:** Writing – review & editing, Methodology. **Alfredo Gay Neto:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization. **Sérgio Frascino Müller de Almeida:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge Petrobras (Petróleo Brasileiro - S.A) for funding scholarships for first and second authors under the agreement number 5900.0112605.19.9. The third author acknowledges National Council for Scientific and Technological Development (CNPq), Brazil for the research grant 304321/2021-4 and The São Paulo Research Foundation (FAPESP) for the research grant 2020/13362-1. The fourth author acknowledges CNPq, Brazil for the research grant 315391/2020-0.

Appendix

Table 8 and Fig. 26 present the results for both the beam and shell models for the 12 load cases at the blade tip.

Table 8
Beam and shell models displacement results for the 12 load cases at the blade tip.

Load case	Model	Unit	Design			Extreme		
			u_x	u_y	u_z	u_x	u_y	u_z
1	Beam	m	3.47	0.59	-0.07	13.77	1.98	-1.56
	Shell	m	3.23	0.64	-0.05	12.85	2.85	-1.31
	Deviation	%	7.43	-7.16	26.73	7.15	-30.68	19.62
2	Beam	m	3.28	5.22	-0.03	15.31	21.31	-5.74
	Shell	m	3.11	5.38	-0.03	15.03	22.01	-6.01
	Deviation	%	5.30	-2.99	1.97	1.83	-3.20	-4.47
3	Beam	m	3.03	10.69	-0.56	11.09	29.91	-8.91
	Shell	m	2.96	10.95	-0.60	11.05	30.39	-9.26
	Deviation	%	2.44	-2.42	-7.29	0.32	-1.60	-3.86
4	Beam	m	1.55	18.03	-2.24	2.93	31.90	-8.98
	Shell	m	1.59	18.49	-2.41	2.91	32.42	-9.40
	Deviation	%	-2.78	-2.49	-7.08	0.60	-1.58	-4.50
5	Beam	m	-1.35	9.12	-0.29	-4.98	26.77	-6.24
	Shell	m	-1.22	9.38	-0.33	-5.70	27.68	-6.84
	Deviation	%	10.77	-2.73	-10.23	-12.65	-3.30	-8.86
6	Beam	m	-2.11	3.21	0.06	-5.28	8.10	-0.42
	Shell	m	-1.92	3.27	0.07	-5.11	8.37	-0.45
	Deviation	%	9.82	-1.64	-13.37	3.39	-3.19	-5.71
7	Beam	m	-2.76	-0.52	-0.10	-5.94	-1.22	-0.40
	Shell	m	-2.57	-0.54	-0.09	-5.88	-1.24	-0.40
	Deviation	%	7.04	-2.44	12.00	0.94	-1.69	0.01
8	Beam	m	-2.61	-4.04	-0.49	-7.26	-10.73	-2.37
	Shell	m	-2.46	-4.15	-0.50	-7.34	-11.09	-2.51
	Deviation	%	6.24	-2.62	-1.65	-1.11	-3.25	-5.73
9	Beam	m	-1.66	-6.15	-0.81	-9.01	-26.82	-10.56
	Shell	m	-1.59	-6.31	-0.85	-9.32	-27.59	-11.16
	Deviation	%	4.39	-2.57	-3.87	-3.34	-2.81	-5.44
10	Beam	m	-0.40	-8.02	-1.20	-0.57	-30.65	-12.69
	Shell	m	-0.42	-8.22	-1.25	-0.46	-31.48	-13.35
	Deviation	%	-4.96	-2.48	-4.41	26.03	-2.63	-4.98
11	Beam	m	2.26	-11.11	-2.11	7.83	-25.55	-9.87
	Shell	m	2.07	-11.42	-2.21	7.77	-26.34	-10.46
	Deviation	%	9.17	-2.71	-4.62	0.73	-3.02	-5.66
12	Beam	m	3.14	-4.69	-0.64	13.06	-16.37	-6.09
	Shell	m	2.89	-4.81	-0.65	12.61	-17.26	-6.47
	Deviation	%	8.88	-2.47	-1.84	3.60	-5.17	-5.87

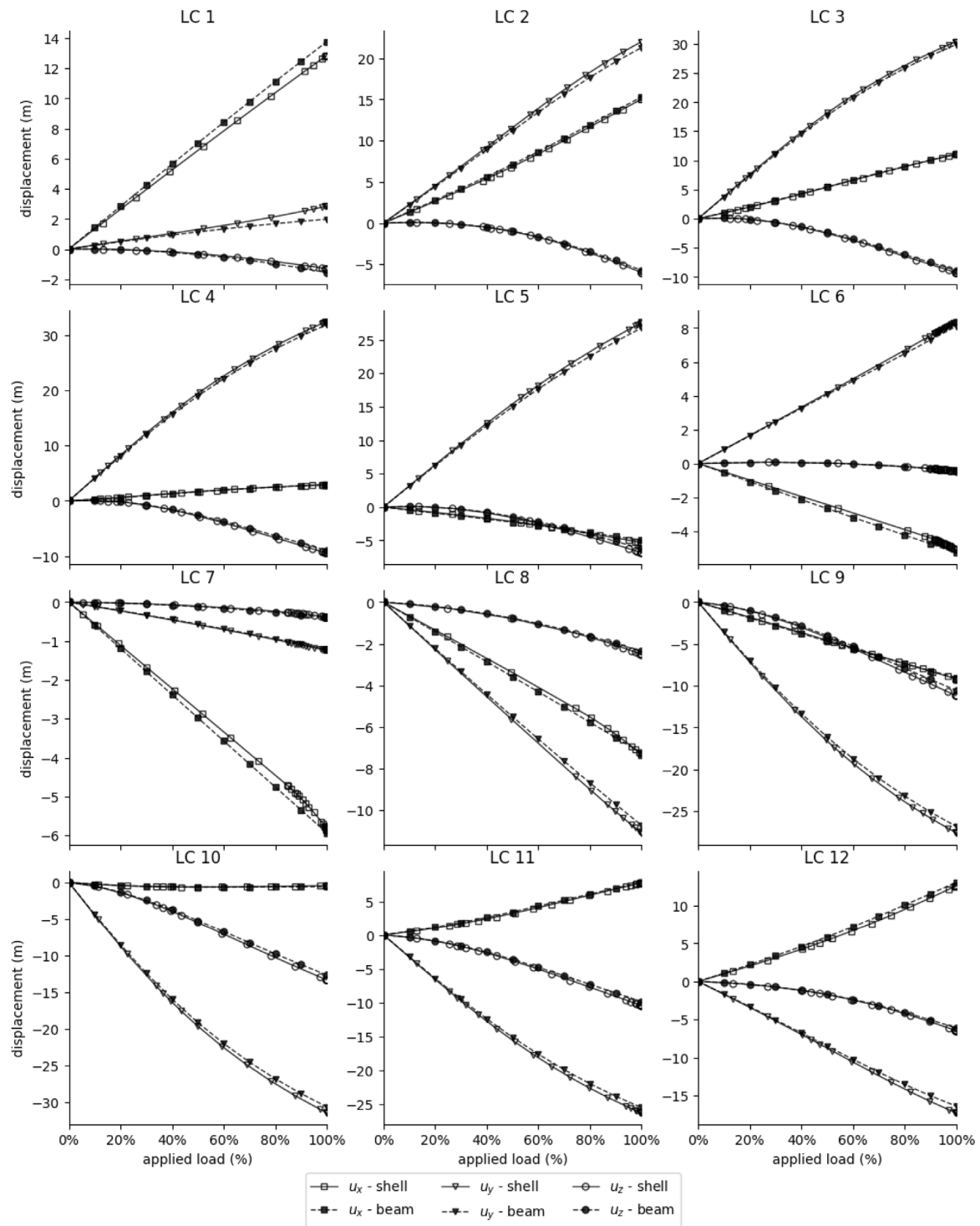


Fig. 26. Load–displacement plots for all load cases. In the figure, 100% of the applied load represents the extreme loading.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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